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ELECTRONIC DEVICE FOR RESTORING IMAGE BY USING INTRINSIC INFORMATION OF INTERMEDIATE LAYER IN MODEL TRAINED TO OUTPUT EXPLICIT INFORMATION AND METHOD THEREOF

Abstract

According to an embodiment, an electronic device receives a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution. The electronic device, based on the received request, executes an image restoration model including an encoder to extract feature information from the first image, a sub-model to determine a text probability map with respect to the first image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information, and a decoder to generate an image of the second resolution, which is connected to the fusion layer. The electronic device provides, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an electronic device for restoring an image by using implicit information of an intermediate layer in a model trained to output explicit information and a method thereof.

BACKGROUND ART

[0002] Technology for processing a photo and/or a video using artificial intelligence is being developed. For example, technology is being developed to classify a subject (e.g., an object including a person, an animal, and/or a vehicle) captured by a photo and/or a video. For example, technology is being developed to recognize one or more characters (or a string) associated with a photo and/or a video.

[0003] The above-described information may be provided as a related art for the purpose of helping understanding of the present disclosure. No argument or decision is made as to whether any of the above description may be applied as a prior art related to the present disclosure.

SUMMARY

Technical Solution

[0004] According to an embodiment, a non-transitory computer readable storage medium storing instructions may be provided. The instructions, when executed by at least one processor of an electronic device individually or collectively, may cause the electronic device to receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on the received request, execute an image restoration model including, an encoder to extract feature information from the first image, a sub-model to determine a text-probability map with respect to the first image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information, and a decoder to generate an image of the second resolution, which is connected to the fusion layer. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.

[0005] According to an embodiment, an electronic device may comprise memory storing instructions, and at least one processor configured to execute the instructions. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on the received request, execute an image restoration model including an encoder to extract feature information from the first image, a sub-model to determine a text-probability map with respect to the first image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information, and a decoder to generate an image of the second resolution, which is connected to the fusion layer. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.

[0006] According to an embodiment, a method of an electronic device may be provided. The method may comprise, based on receiving an image, obtaining a sub-model trained to output a text-probability map indicating one or more characters associated with the image. The method may comprise performing, using the sub-model, training of an image restoration model including an encoder to extract feature information from an input image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, prior to an output layer of the sub-model which receives the input image, and the feature information, and a decoder, that is connected to the fusion layer, to generate an output image having a second resolution greater than a first resolution of the input image. The method may comprise providing the image restoration model as a portion of a software application to restore the image.

[0007] According to an embodiment, an electronic device may comprise memory storing instructions, and at least one processor configured to execute the instructions. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on receiving an image, obtain a sub-model trained to output a text-probability map indicating one or more characters associated with the image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to perform, using the sub-model, training of an image restoration model including an encoder to extract feature information from an input image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, prior to an output layer of the sub-model which receives the input image, and the feature information, and a decoder, that is connected to the fusion layer, to generate an output image having a second resolution greater than a first resolution of the input image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide the image restoration model as a portion of a software application to restore the image.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** illustrates an exemplary block diagram of an electronic device for restoring at least a portion of an image.

[0009] FIG. **2** illustrates an exemplary block diagram of a structure of an image restoration model executed by an electronic device according to an embodiment.

[0010] FIG. **3** illustrates an exemplary block diagram of a structure of a sub-model in an image restoration model trained to output a text probability map.

[0011] FIG. **4** illustrates an exemplary block diagram of a structure of an image restoration model executed by an electronic device according to an embodiment.

[0012] FIG. **5** illustrates an exemplary block diagram of an image restoration model connected to a teacher model.

[0013] FIG. **6** illustrates graphs for describing a performance of an electronic device executing an image restoration model according to an embodiment.

[0014] FIG. **7** illustrates graphs for describing a performance of an electronic device executing an image restoration model according to an embodiment.

[0015] FIGS. **8**A and **8**B illustrate at least one license plate (or number plate), which is a subject included in an image restored by an image restoration model, according to an embodiment.

[0016] FIG. **9** is a diagram for describing an overconfidence phenomenon.

DETAILED DESCRIPTIONS

[0017] Hereinafter, various embodiments of the present document will be described with reference to the accompanying drawings.

[0018] FIG. **1** illustrates an exemplary block diagram of an electronic device **101** to restore at least a portion of an image **150**. The electronic device **101** may be configured to at least partially restore or enhance the image **150**. Restoring or enhancing the image **150** may include an operation of improving visibility of a subject represented by the image **150** by compensating for distortion included in the image **150**, such as blur, afterimage, and optical flow.

[0019] Referring to FIG. **1**, the image **150** including a portion **152** associated with a license plate (or a number plate) is exemplarily illustrated. For example, the image **150** may be transmitted from an external electronic device to the electronic device **101** through communication circuitry **130**. For example, the image **150** may be obtained using a camera **140** included in the electronic device **101**. For example, the image **150** may be a file with a format based on a form compressing and storing an image such as a joint photographic experts group (jpeg), a Portable Network Graphics (PNG). For example, the image **150** may include raw data obtained from the camera **140**. For example, the image **150** may be included in a sequence (e.g., a video) of image frames, which is included in a video and set to be displayed sequentially. A means for obtaining or receiving the image **150** is not limited to the communication circuitry **130** and/or the camera **140** illustrated in FIG. **1**.

[0020] Referring to the exemplary image **150** of FIG. **1**, an exemplary subject such as a vehicle may be captured. The image **150** may be distorted according to an environment in which a subject is photographed. For example, in case that the subject is moving (e.g., driving of a vehicle), and/or a camera (e.g., the camera **140**) controlled to obtain the image **150** is moving (or shaking), an appearance of the subject represented by pixels of the image **150** may be distorted. According to an embodiment, the electronic device **101** may enable the appearance of the subject represented by the image **150** to be clear, by at least partially reducing or removing the distortion generated in the image **150**.

[0021] Referring to FIG. **1**, an exemplary hardware configuration of the electronic device **101** to at least partially restore the image **150** is illustrated. For example, the electronic devices **101** may include a personal computer such as a laptop and a desktop, a smartphone, a smart pad, and a tablet PC. For example, the electronic device **101** may include a smart accessory such as a smartwatch, a smart ring, and/or a head-mounted device (HMD). For example, the electronic device **101** may be referred to as a mobile device, user equipment (UE), a multifunction device, a portable communication device, and/or a portable device. For example, the electronic device **101** may be included as an electronic control unit (ECU) in a vehicle (e.g., an electric vehicle (EV)). For example, the electronic device **101** may include a server of a service provider that provides a service for restoring the image **150**. The server may include one or more PCs and/or workstations.

[0022] Referring to FIG. **1**, according to an embodiment, the electronic device **101** may include at least one of a processor **110**, memory **120**, the communication circuitry **130**, or the camera **140**. According to an embodiment, the communication circuitry **130** and/or the camera **140** may not be included in the electronic device **101**. For example, the communication circuitry **130** and/or the camera **140** may be disposed outside the electronic device **101** and may be electrically connected to the electronic device **101**.

[0023] Referring to FIG. **1**, the processor **110**, the memory **120**, the communication circuitry **130**, and the camera **140** may be electronically and/or operably coupled with each other by an electronical component such as a communication bus **102**. Hereinafter, electronical components being operably combined may mean that a direct connection or an indirect connection between first electronical components and second electronical components is established by wire or wirelessly so that a second electronical component is controlled by a first electronical component. Although illustrated based on different blocks, an embodiment is not limited thereto, and a portion of (e.g., at least a portion of the processor **110**, the memory **120**, and the communication circuitry **130**) the electronical components of FIG. **1** may be included in a single integrated circuit such as a system on a chip (SoC). A type and/or the number of electronical components included in the electronic device **101** is not limited as illustrated in FIG. **1**. For example, the electronic device **101** may include only a portion of the electronical components illustrated in FIG. **1**.

[0024] The processor **110** of the electronic device **101** according to an embodiment may include circuitry (e.g., processing circuitry) for processing data based on one or more instructions. The circuitry for processing data may include, for example, an arithmetic and logic unit (ALU), a floating point unit (FPU), a field programmable gate array (FPGA), a central processing unit (CPU), a graphic processing unit (GPU), a neural processing unit (NPU), and/or an application processor (AP). For example, the number of the processors **110** may be one or more. The processing circuitry of the processor **110** that loads (or fetches) an instruction and performs a calculation corresponding to the loaded instruction may be referred to or referenced as core circuitry (or a core). For example, the processor **110** may have a structure of a multi-core processor including a plurality of core circuitries, such as a dual core, a quad core, a hexa core, or an octa core. A function and/or an operation described with reference to the present disclosure may be individually and/or collectively performed by one or more processing circuitries included in the processor **110**.

[0025] According to an embodiment, the memory **120** of the electronic device **101** may include circuitry for storing data and/or an instruction inputted and/or outputted to the processor **110**. The memory **120** may include, for example, volatile memory such as random-access memory (RAM) and/or non-volatile memory such as read-only memory (ROM). The non-volatile memory may be referred to as storage. The volatile memory may include, for example, at least one of dynamic RAM (DRAM), static RAM (SRAM), cache RAM, and pseudo SRAM (PSRAM). The non-volatile memory may include, for example, at least one of programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, a hard disk, a compact disk, a solid state drive (SSD), and an embedded multi media card (eMMC). The memory **120** may include one or more storage mediums (e.g., the volatile memory and/or nonvolatile memory described above) positioned in the electronic device **101** in a distributed manner. The processor **110** of the electronic device **101** may perform a function and/or an operation indicated by instructions, by executing the instructions of the memory **120** in the electronic device **101**. For example, in case that the electronic device **101** includes at least one processor, the at least one processor may be configured to execute the instructions collectively or individually.

[0026] According to an embodiment, the communication circuitry **130** of the electronic device **101** may include hardware for supporting transmission and/or reception of an electrical signal between the electronic device **101** and the external electronic device (e.g., a user terminal configured to transmit the image **150**). The communication circuitry **130** may include at least one of, for example, a modem, an antenna, and an optic/electronic (O/E) converter. The communication circuitry **130** may support transmission and/or reception of an electrical signal based on various types of protocols, such as Ethernet, a local area network (LAN), a wide area network (WAN), wireless fidelity (WiFi), near field communication (NFC), Bluetooth, bluetooth low energy (BLE), ZigBee, long term evolution (LTE), fifth generation (5G), a new radio (NR), sixth generation (6G), and/or above-6G.

[0027] According to an embodiment, the camera **140** of the electronic device **101** may include one or more optical sensors (e.g., a charged coupled device (CCD) sensor and a complementary metal oxide semiconductor (CMOS) sensor) that generate an electrical signal indicating a color and/or brightness of light. The plurality of optical sensors included in the camera **140** may be disposed in a form of a 2 dimensional array. The camera **140** may generate 2 dimensional frame data corresponding to light reaching the optical sensors of the 2 dimensional array, by obtaining an electrical signal of each of the plurality of optical sensors substantially simultaneously. For example, photo data captured using the camera **140** may mean a 2 dimensional frame data obtained from the camera **140**. For example, video data captured using the camera **140** may mean a sequence of a plurality of 2 dimensional frame data obtained from the camera **140**.

[0028] Referring to FIG. **1**, the processor **110** of the electronic device **101** according to an embodiment may at least partially restore or enhance the image **150** by executing an image restoration program **125**. The processor **110** (e.g., the CPU, the GPU, and/or the NPU) executing the image restoration program **125** may perform calculations for restoring the image **150**. The calculations may be associated with a computational model (e.g., an artificial neural network, and/or a neural network) configured to simulate a neural activity of a living organism. The neural activity may include, for example, a cognitive activity, an inference activity, and/or a creative activity of a living organism. For example, instructions indicating the computational model, formulas associated with the computational model, and/or a constant (e.g., coefficients and/or weights) included in the formulas, may be at least partially included in

the image restoration program **125**.
[0029] According to an embodiment, the processor **110** of the electronic device **101** may restore or enhance the portion **152** (e.g., a portion of an object in which one or more characters are printed is captured, such as a number plate and/or a sign plate) in which at least one character is captured, in the image **150**. For example, in the image **150**, the electronic device **101** may extract or segment (or crop) the portion **152** associated with at least one character. The portion **152** may be referred to as a region of interest (ROI). The processor **110** may restore or enhance the portion **152** by executing the image restoration program **125**.
[0030] In an embodiment, the electronic device **101** may increase or enhance a resolution of a scene by recognizing text (e.g., text that is indicated as being captured or included in the scene) associated with the scene such as the image **150**. For example, in case of detecting one or more characters from a scene of a relatively low resolution (or small size), the electronic device **101** may generate another scene corresponding to the scene and having a higher resolution (or a larger size) than the resolution of the scene, by using a shape and/or an appearance of the detected one or more characters. For example, with respect to a scaling factor f, from a scene with a width w and a height h, the electronic device **101** may generate or output a scene with a width fw and a height fh.
[0031] In an embodiment, in terms of recognizing text and generating a high-resolution scene, the image restoration program **125** and/or artificial intelligence driven by the image restoration program **125** may be referred to as a scene text image super-resolution (STISR) and/or a model for the STISR. A performance of the STISR may be evaluated using accuracy (e.g., STISR accuracy) of a character included in the high-resolution scene generated by executing the STISR.
[0032] Referring to FIG. **1**, an image **160** that the electronic device **101** outputs as a result of restoring the portion **152** of the image **150** is illustrated. The image **150** and/or the portion **152** may be referred to as an input image in terms of being inputted to the processor **110** of the electronic device **101**. The image **160** may be referred to as an output image in terms of output data corresponding to the input image. According to an embodiment, the electronic device **101** may obtain information indicating one or more characters associated with the portion **152** by using an artificial intelligence model trained to recognize one or more characters from an image. By using the information, the electronic device **101** may generate or output the image **160** as a high-resolution image corresponding to the portion **152**.
[0033] Referring to FIG. **1**, the image **160** may have a larger size than the portion **152** and/or a higher resolution than the portion **152**. Dimensions (e.g., a width and/or a height) of the image **160** may be greater than dimensions of the portion **152**. For example, the image **160** may have the same dimensions and/or resolution as the image **150**. In an embodiment of receiving the image **150** and/or the portion **152** from the external electronic device through the communication circuitry **130**, the electronic device **101** may receive a request for restoring the portion **152** of the image **150** with a first resolution to the image **160** with a second resolution greater than the first resolution. From a signal received from the external electronic device, the electronic device **101** may identify or detect the image **150** and/or the portion **152**. The signal may include a command and/or an operand indicating the request for restoration of the portion **152**. In an embodiment of receiving the entire image **150** including the portion **152**, the processor **110** of the electronic device **101** may extract or segment the portion **152** in which a subject relation to one or more characters is captured, such as a number plate. The portion **152** may be used as an image used for restoration.
[0034] Based on the request for restoring the image **150** and/or the portion **152**, the electronic device **101** may execute an artificial intelligence model (e.g., an image restoration model) provided by the image restoration program **125**. The electronic device **101** may provide the image **160** of the second resolution, obtained based on the execution of the image restoration model, as a response to the request. For example, the electronic device **101** may transmit a signal including the image **160** to the external electronic device through the communication circuitry **130**.
[0035] In an embodiment, the image restoration model executed by the image restoration program **125** may include a sub-model trained to recognize one or more characters (e.g., indicated to be captured by an input image) associated with the input image (e.g., the portion **152** and/or the image **150** including the portion **152**) inputted to the image restoration model. The sub-model, which is information (e.g., explicit information) readable by the processor **110** executing a software application distinct from the image restoration model and/or the image restoration program **125**, may be trained to output information indicating the one or more characters associated with the input image, degrees to which each of the one or more characters is associated with the input image (e.g., probabilities that one or more characters are captured by the input image), and/or a positional relationship of the one or more characters (e.g., a position and/or an order of each of the one or more characters in a string).
[0036] For example, the information outputted from the sub-model may be referred to as text probability information in terms of including probabilities indicating text indicated to be captured by the input image. The text probability information may be referred to as text categorical information, text probability, a text probability map, text prior information, and/or text distribution. For example, the text probability information may include category information of text and/or information indicating a visual cue for text in an image.
[0037] According to an embodiment, the electronic device **101** may be trained to generate the image **160** using an

intermediate state and/or intermediate information of the sub-model trained to output explicit information such as the text probability information. For example, among nodes (e.g., perceptrons) of the sub-model, which are distinguished by a plurality of layers, values of nodes that are different from nodes of an output layer including nodes corresponding to each element of the text probability information may be directly transmitted to another sub-model of the image restoration model. For example, an intermediate layer of the sub-model may be connected to the other sub-model of the image restoration model.

[0038] For example, values of nodes included in the intermediate layer may be implicit information that is distinct from explicit information. The implicit information may include more detailed information with respect to an input image than text probability information, which includes only probabilities that the input image (e.g., the portion **152** and/or the image **150**) corresponds to each of a plurality of characters. By executing the image restoration model using the implicit information, the electronic device **101** may restore the portion **152** more accurately. For example, the electronic device **101** may obtain or generate the image **160** that more accurately represents one or more characters included in the portion **152**. In the example, since more accurately recognizing or representing one or more characters from the portion **152**, when receiving requests to repeatedly restore the portion **152**, a plurality of images (e.g., the image **160**) generated in response to the requests may include similar characters to each other.

[0039] Hereinafter, an exemplary structure of the image restoration model executed by the image restoration program **125** and a process of training the image restoration model will be exemplarily described with reference to FIGS. **2** to **5**.

[0040] FIG. **2** illustrates an exemplary block diagram of a structure of an image restoration model executed by an electronic device (e.g., the electronic device **101** of FIG. **1**) according to an embodiment. The electronic device **101** and/or the processor **110** of FIG. **1** may execute the image restoration model described with reference to FIG. **2** by executing an image restoration program **125**.

[0041] Hereinafter, an operation of executing an artificial intelligence model, such as the image restoration model, may include operations of performing one or more calculations associated with the artificial intelligence model by using a processor device (e.g., the processor **110** of FIG. **1** including the GPU and/or the NPU) of the electronic device. The operation of executing the artificial intelligence model may include an operation of inputting commands (or instructions) indicating the calculations to the GPU and/or the NPU to perform the calculations by the GPU and/or the NPU. The operation of executing the artificial intelligence model may include an operation of inputting data (e.g., the input image such as the image **150** and/or the portion **152** of FIG. **1**) to be at least partially changed by the calculations to the GPU and/or the NPU. Although the operation of executing the artificial intelligence model based on the GPU and/or the NPU has been exemplarily described, an embodiment is not limited thereto, and an operation of executing the artificial intelligence model using a CPU may also be performed similarly to the above-described operation.

[0042] Referring to FIG. **2**, calculations performed by the image restoration model are illustrated as a plurality of blocks for distinguishing types and/or an order of the calculations. Any one block of FIG. **2** may correspond to a group of calculations performed while executing the artificial intelligence model (e.g., the image restoration model). Each of the blocks of FIG. **2** may be referred to as a computation, layer(s), a sub-model and/or a module for the artificial intelligence model. Referring to FIG. **2**, the image restoration model including a teacher model **210** connected to the image restoration model is exemplarily illustrated to train at least a portion of the image restoration model.

[0043] For example, the image restoration model may include an encoder (e.g., a combination of a spatial transformer networks (STN) computation **241** and a convolution computation **242**) for extracting feature information from an image. The encoder including the STN calculation **241** and/or the convolution calculation **242** may include a shallow convolutional neural network (CNN) that has a small loss of structural information (or spatial information) required for restoring the image. The encoder (or a STISR) of the image restoration model may include a relatively small number of layers to reduce the loss of the structural information (or the spatial information) of a low-resolution image when extracting a feature of the low-resolution image to perform a low-level vision task (e.g., a task of increasing a resolution of an image). By executing the encoder of the image restoration model, the electronic device may generate or obtain feature information on an input image **202**. The feature information may include summarized (or reduced dimensional) information of the input image **202** to specify or distinguish the input image **202**. The feature information may include positions and/or characteristics of one or more pixels uniquely included in the input image **202**, such as a feature point (or a key point) and/or a boundary line.

[0044] For example, the image restoration model may include a sub-model **220** for determining a text probability map with respect to the input image **202**. The teacher model **210** may generate training information (e.g., ground truth data and input data corresponding to the ground truth data) used to train the sub-model **220** using knowledge distillation. The number of calculations of the sub-model **220** and parameters (e.g., coefficients and/or weights) used in the calculations, may be less than the number of calculations of the teacher model **210** and parameters used in the calculations of the teacher model **210**. For example, the sub-model **220** may be pre-trained by the teacher model **210** executed using the parameters more than the parameters for the sub-model **220**.

[0045] In an embodiment, the teacher model **210** used for training the sub-model **220** may be trained to recognize one or more characters from a scene such as an image **201**. In terms of character recognition, the teacher model **210** may be referred to as a scene-text recognizer (STR) and/or a STR model. The teacher model **210** may be configured to recognize or process a feature such as a shape and/or a position of the one or more characters in the image **201**.

[0046] Referring to FIG. **2**, types and orders of calculations of the teacher model **210** and the sub model **220** may be similar or identical to each other. For example, when executing the sub-model **220**, the electronic device may obtain or generate output data (e.g., text probability information and/or the text probability map) by sequentially performing an encoding computation **220***a*, a sequence modeling computation **220***b*, a decoding prediction computation **220***c*, and a linearization computation **220***d* on the input image **202**. The computations (e.g., the encoding computation **220***a*, the sequence modeling computation **220***b*, the decoding prediction computation **220***c*, and the linearization computation **220***d*) sequentially performed in the sub-model **220** may correspond to computations (e.g., an encoding computation **210***a*, a sequence modeling computation **210***b*, a decoding prediction computation **210***c*, and a linearization computation **210***d*) sequentially performed in the teacher model **210**, respectively. A connection of the computations described above may have a structure of thin plate spline transformation (TPS)-Residual neural Network (ResNet)-bidirectional long-short term memory (BiLSTM)-attention mechanism (TRBA). An exemplary structure of the sub-model **220** having a structure of the TRBA will be described in detail with reference to FIG. **3**. An embodiment is not limited to thereto, and another structure (or a topology) such as a convolution-recurrent neural network (CRNN), an autonomous, bidirectional and iterative network (ABINet), and/or a permuted autoregressive sequence (PARseq) may be applied to the structure of the sub-model **220**. An output layer of the sub-model **220** may include values determined by calculations performed for a linearization computation **220***d*. The values included in the output layer may be the text probability information.

[0047] According to an embodiment, the electronic device may train the sub-model **220** using the teacher model **210** to which the image **201** having a relatively high resolution is inputted. For example, the electronic device executing the teacher model **210** may determine, from the image **201**, the text probability map indicating one or more characters associated with the image **201**. The electronic device may train the sub-model **220** using another image having a lower resolution than the image **201** and the determined text probability map.

[0048] Referring to FIG. **2**, the output layer of the sub-model **220** may be associated with the linear computation **220***d*. In the sub-model **220**, implicit information including a result of performing the decoding prediction computation **220***c* (or a state of any one intermediate layer for the decoding prediction computation **220***c*) and to be used in the linear computation **220***d* may be provided to a fusion layer **243**. Prior to being provided to the fusion layer **243**, implicit information may be inputted to a projection model **230**. Using the projection model **230**, the electronic device may sequentially perform a linear computation **232***a*, a Parametric rectified linear unit (ReLU) (PReLU) computation **232***b* (e.g., computations included in the sub-model **232**), and a prior interpreter computation **234** for the implicit information. Implicit information that is at least partially changed by the projection model **230** may be inputted to the fusion layer **243**. The projection model **230** may be referred to as a Non-CAtegorical Prior (NCAP) in terms of outputting implicit information (e.g., non-categorical information) of the sub-model **220** trained to generate categorical information. A combination of the sub-model **220** and the projection model **230** may be referred to as a scene-text recognizer (STR). Information outputted by the projection model **230** (e.g., information transmitted from the projection model **230** to the fusion layer **243**) may be referred to as prior knowledge information.

[0049] The combination of the sub-model **220** and the projection model **230** may cause the electronic device executing the image restoration model to generate the output image **203** using textual information (e.g., the text probability information) inferred from the input image **202**. The encoder, which is a combination of the spatial transformer networks (STN) computation **241** and the convolution computation **242**, may cause the electronic device executing the image restoration model to generate the output image **203** using nontextual information (e.g., the structural information) inferred from the input image **202**. In terms of both the textual information and the nontextual information being used, the image restoration model may be a model supporting multimodality.

[0050] Referring to FIG. **2**, the fusion layer **243** may be configured to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output a text probability map and the feature information. For example, the electronic device may perform calculations indicated by the fusion layer **243** by using both feature information including a result of performing the convolution computation **242** of the encoder and the text probability map outputted or generated from the sub-model **220** and/or the projection model **230**.

[0051] Referring to FIG. **2**, the image restoration model may perform a decoder computation **244** to generate the output image **203** having a resolution higher than a resolution of the input image **202** by using information generated by the fusion layer **243**. The decoder computation **244** may be trained to generate the output image **203** that has a resolution greater than the input image **202** and/or a size wider than the input image **202**, and that is associated with the input image **202** (e.g., including content of the input image **202**) by using the information generated by the fusion layer **243**. The output image **203** may be provided as a result of restoring or enhancing the input image **202**.

[0052] As described above, according to an embodiment, the electronic device may generate or obtain the output image **203** from the input image **202** by executing the image restoration model including the sub-model **220** trained to output the text probability map indicating one or more characters indicated as being captured by the input image **202**, and positions of the one or more characters. The image restoration model may include the fusion layer **243** connected (e.g., indirectly connected through the projection model **230**) to the intermediate layer (e.g., an intermediate layer to perform the decoding prediction computation **220***c*) of the sub-model **220** to extract the implicit information used to determine the text probability map, which is explicit information. For example, in order to reduce or prevent distortion of the output image **203** due to an error (e.g., a result of incorrectly recognizing at least one character from the input image **202**) that may be included in the text probability map, the electronic device may fuse or generate the output image **203** by using the implicit information, which is used to determine the text probability map and includes various information on the input image **202** compared to the text probability map.

[0053] In an embodiment, since the implicit information includes higher-dimensional information compared to the text probability map, the electronic device may effectively resolve a domain gap due to a resolution difference between the input image **202** and the output image **203**. For example, without a domain transfer, the electronic device may obtain or generate information (e.g., the implicit information) to be used to reduce or remove the domain gap.

[0054] In an embodiment, after the sub-model **220** included in the image restoration model to restore the output image **203** from the input image **202** is trained by the teacher model **210**, the sub-model **220** configured to obtain information (e.g., the text probability map) on one or more characters may be retrained. The retrained sub-model **220** may generate or output a feature (e.g., a discriminative feature) useful to remaining layers (e.g., the projection model **230**, the fusion layer **243**, and/or the decoder **244**) of the image restoration model that is executed by generating the output image **203** connected to the sub-model **220**. The image restoration model including the retrained sub-model **220** may be trained using ground truth data (e.g., a pair of the output image **203** and the input image **202** obtained by distorting the output image **203** and having a smaller resolution and/or a smaller size than the output image **203**).

[0055] For example, the image restoration model may be trained to output the output image **203** as a result of enhancing the input image **202** by a training process of a first step of retraining the pre-trained sub-model **220** and a second step of training the image restoration model including the retrained sub-model **220**. The first step of the training process will be described with reference to FIG. **3**. The second step of the training process will be described with reference to FIG. **4**.

[0056] FIG. **3** illustrates an exemplary block diagram of a structure of a sub-model **220** in an image restoration model trained to output a text probability map. The electronic device **101** and/or the processor **110** of FIG. **1** may obtain or execute the sub-model **220** and/or the image restoration model including the sub-model **220** described with reference to FIG. **3**, by executing an image restoration program **125**.

[0057] According to an embodiment, based on receiving an image, the electronic device may obtain the sub-model **220** trained to output a text probability map indicating one or more characters associated with the image. The electronic device may perform (e.g., fine-tuning) training again on the obtained sub-model **220** using a loss function. The loss function may be set or defined to generate explicit information (e.g., text probability information) outputted from the sub-model **220** as well as implicit information indicating a discriminative feature to be used by the image restoration model including the sub-model **220**.

[0058] Referring to FIG. **3**, an exemplary structure of the sub-model **220** having a structure of a TRBA is illustrated. Based on the structure of the TRBA, the sub-model **220** may include a chain connection of a TPS model **310** for a TPS computation, a backbone model **320** (e.g., ResNET), a BiLSTM **330**, a first feed-forward model **340**, a plurality of RNN decoders **350**, and a second feed-forward model **360**. The electronic device **101** may input an image **301** (x∈custom-character.sup.h×w×c) as an input layer of the TPS model **310**. x of the x∈custom-character.sup.h×w×c may mean an input image (e.g., the image **301**) inputted to an input layer. For example, the image **301** having a size of a height h and a width w and having c number of channels (e.g., three channels each of red, green, and blue) may be indicated.

[0059] In an embodiment, the sub-model **220** may be trained (e.g., pre-trained) to output explicit information p of Equation 1.

[00001] $p(y_{hr}) = STR(x_{HR})$ [Equation1]

[0060] The explicit information p of Equation 1 may include output data (e.g., P0, P1, . . . , Pt) of the second feed-forward model **360** of FIG. **3**. The output data Pt (herein, 0≤t≤1) of the second feed-forward model **360** may be determined based on output data h=[h1, h2, . . . , hi] from a first timing (or a first time step) to a t-th timing (or a t-th time step) of the RNN decoders **350**. The explicit information p of Equation 1 may be determined based on a projection of the output data h=[h1, h2, . . . , hi] of the RNN decoders **350**. For example, output data of the sub-model **220** may be set as in Equation 2.

[00002] $t_{LR} = p(\tilde{y}_i \text{ .Math. } \{(\tilde{y}_1, \text{ .Math. }, \tilde{y}_{i-1}, x) = g(h_i)\})$ [Equation2]

[0061] h.sub.i of Equation 2 may be an intermediate state vector of an intermediate layer (e.g., the RNN decoders **350**) of the sub-model **220** of the t-th timing (or the t-th time step).

[0062] In an embodiment, the sub-model **220** may be trained by a loss function that increases and/or maximizes a difference and/or a margin of probabilities between classes determined by the sub-model **220**, as well as cross-entropy loss. The loss function may be defined to generate a discriminative feature for classes (e.g., classes corresponding to each of a plurality of characters) of the sub-model **220** and/or to alleviate confusion between the classes. The loss function may be used to retrain the pre-trained sub-model **220** to output the explicit information p from the image **301**.

[0063] For example, a loss function custom-character.sub.str set to retrain the sub-model **220** may be set to increase or maximize a distance and/or a margin from a decision boundary surface. For example, it may be defined to maximize the margin of the decision boundary surface between a specific class y.sub.i and another class y.sub.j,j≠i. For example, the loss function custom-character.sub.str may be defined as a sum custom-character.sub.str=custom-character.sub.rec +custom-character.sub.aux between custom-character.sub.rec of Equation 3 and custom-character.sub.aux of Equation 4.

[00003] $\mathcal{L}_{\text{rec}} = -\underset{t=0}{\overset{l}{\text{.Math.}}}\ \underset{i=1}{\overset{\text{.Math. } A \text{ .Math.}}{\text{.Math.}}}\ y_{t,i} \log p_{t,i}$ [Equation3]

$\mathcal{L}_{aux} = -\min(1, \underset{t=1}{\overset{l}{\text{.Math.}}}\ \underset{i=1, i \neq p_{y_t}}{\overset{|A|}{\text{.Math.}}}\ \log(p_{y_t} - p_{t,i} + \quad))$ [Equation4]

[0064] custom-character.sub.rec of Equation 3 is a cross entropy for training the sub-model **220**, and p.sub.t,i may indicate a probability of being matched to a i-th class among classes (e.g., characters) that may be distinguished by the sub-model **220**. |A| of Equation 3 may indicate the total number of classes. 1 of Equation 3 may correspond to the number of RNN decoders **350**.

[0065] custom-character.sub.aux of Equation 4 may be defined to obtain a discriminative feature (or implicit information) using the sub-model **220**. When training the sub-model **220** using the image **301** and truth data indicating one or more characters included in the image **301**, p.sub.y.sub.t of Equation 4 may indicate an output probability of the sub-model **220** corresponding to a character indicated as a correct answer by the truth data. Referring to Equation 3, since i does not match the character indicated as the correct answer (i≠p.sub.y.sub.t), p.sub.t,i may indicate the output probability of the sub-model **220** corresponding to character that is not the correct answer. ϵ of Equation 4 may be a real number (e.g., 10.sup.−7) defined so that a result value of a log function is not reduced to a too small value (e.g., negative infinity). 1 of Equations 3 and 4 may indicate a length of a maximum character string, and i of Equations 3 and 4 may be a variable that changes in the total number |A| of the classes.

[0066] Using a loss function (e.g., custom-character.sub.str=custom-character.sub.rec+custom-character.sub.aux) for a difference between the output data and the truth data of the sub-model **220** for the image **301**, the sub-model **220** may be trained so that the output data h=[h1, h2, . . . , hi] of the RNN decoders **350** has a discriminative feature. The output data h=[h1, h2, . . . , hi] of the RNN decoders **350** may be used for execution and/or training of the image restoration model including the sub-model **220**. Hereinafter, an exemplary structure of the image restoration model using the output data h=[h1, h2, . . . , hi] outputted from the sub-model **220** as implicit information of the sub-model **220** will be described with reference to FIG. **3**.

[0067] FIG. **4** illustrates an exemplary block diagram of a structure of an image restoration model executed by an electronic device according to an embodiment. The electronic device **101** and/or the processor **110** of FIG. **1** may execute or train the image restoration model described with reference to FIG. **4** by executing an image restoration program **125**.

[0068] Referring to FIG. **4**, the image restoration model may include a TPS model **420** and a shallow CNN **421**. The electronic device **101** may extract low-level feature information by performing calculations indicated by the TPS model **420** and the shallow CNN **421**, from an input image **402**. By combining the feature information with position embedding data for a fusion operation, the electronic device **101** may obtain feature information of F.sub.v∈custom-character.sup.c×hw. C of custom-character.sup.c×hw, which is a number indicating a dimension of feature information, may correspond to the number of dimensions of information outputted from an output layer of the shallow CNN **421**. hw of custom-character.sup.c×hw may indicate a size (e.g., the number of parameters arranged in one dimension) of flattened information (e.g., one-dimensional information) of the input image **402**.

[0069] By performing calculations indicated by the TPS model **420**, the electronic device **101** may adjust shapes of characters within the input image **402** so that the characters have uniform shapes. For example, information outputted from a Flatten model **422** connected to the shallow CNN **421** may correspond to F.sub.v of Equation 5.

[00004] $F_v = \text{Flatten}(\text{Enc}_1(TPS(x_{LR})) + PE)$ [Equation5]

[0070] x.sub.LR of Equation 5 may indicate the input image **402** having a relatively low resolution. PE of Equation 5 may indicate position embedding data combined with feature information. Flatten of Equation 5 may indicate a computation of converting multidimensional information into one-dimensional information. Enc.sub.1 of Equation 5 may indicate a computation performed in the shallow CNN **421**. According to an embodiment, the image restoration model may be trained to use information (e.g., the position embedding data PE of Equation 5) indicating a spatial characteristic of an image to consider a distance between pixels within the image while calculating feature information.

[0071] In a state of processing the input image **402** using the image restoration model, the electronic device **101** may perform a first operation of processing the input image **402** using the TPS **420** and/or the shallow CNN **421** and a second operation of processing the input image **402** using the sub-model **220** in parallel (or substantially simultaneously). The first operation and the second operation may be performed substantially simultaneously by different processors included in the electronic device **101**. By using the sub-model **220** in a state trained based on the operation described with reference to FIG. **3**, the electronic device **101** may obtain implicit information P.sub.NCAP∈custom-character.sup.l×embed from the input image **402**. Implicit information P.sub.NCAP may be determined or may be calculated, based on Equation 6.

[00005]

$$p_{\text{NCAP}} \in \mathbb{R}^{l\times embed} = PReLU(STR_{\text{stu},dec}(STR_{\text{stu},enc}(x_{LR})) \text{ .Math. } W_{proj}) = \text{PReL}U(h \text{ .Math. } W_{proj}) \quad \text{[Equation6]}$$

[0072] A STR term of Equation 6 may mean a scene text recognizer, and STR.sub.str,enc may indicate a computation performed in a decoder (e.g., a group of BiLSTM, Attention mechanism, and Linear in the sub-model **220**) of the sub-model **220**. STR.sub.str,enc may indicate a computation performed by an encoder (e.g., ResNet in the sub-model **220**) of the sub-model **220**. X.sub.LR of Equation 6 may indicate the input image **402** having a relatively low resolution.

[0073] By using information (e.g., P.sub.NCAP of Equation 6) obtained from a NCAP projector **410**, the electronic device may obtain, or calculate, feature information F.sub.p of Equation 7 from a projection model **230**.

[00006][Equation7]$F_p = (p_{\text{NCAP}} + \text{PE}) \text{ .Math. } W_p$

[0074] By performing a softmax computation and/or a layer normalization computation on feature information obtained from the projection model **230**, the electronic device may obtain or calculate feature information F.sub.p′ of Equation 8.

[00007][Equation8]$F_p' = \text{LN}(\text{softmax}(\frac{Q_p K_p^T}{\sqrt{d}})V_p + F_p)$

[0075] From the feature information F.sub.p and F.sub.p′ of Equations 7 and 8, the electronic device may obtain or calculate feature information F.sub.p″ of Equation 9.

[00008][Equation9]$F_p'' = LN(F_p' \text{ .Math. } W_p' + F_p')$

[0076] Equation 9 may correspond to self-attention of F.sub.p′ of Equation 8. For the self-attention, for example, Equation 9 may be defined to process the feature information F.sub.p′ of Equation 8 using a projection and a linear computation (LN) based on an fc layer. An addition computation (e.g., +F.sub.p computation and/or +F′.sub.p computation) of Equation 8 and Equation 9 may indicate a residual connection (or identity mapping).

[0077] According to an embodiment, the electronic device may process the implicit information obtained from the sub-model **220** using the projection model **230**. In the projection model **230**, the NCAP projector **410**, a multi-head self-attention model **411**, a first layer normalization model **412**, a feed forward model **413**, and a second layer normalization model **414** may be combined in a chain. Using the projection model **230**, the electronic device may generate or obtain feature information F.sub.p∈custom-character.sup.l×proj from the implicit information. The feature information generated by the projection model **230** may include non-categorical information recognized by the sub-model **220** from the input image **402**.

[0078] According to an embodiment, the electronic device may perform multi-head cross-attention between the feature information F.sub.v∈custom-character.sup.c×hw of the shallow CNN **421** and the feature information F.sub.p∈custom-character.sup.l×proj of the projection model **230** in a multi-head cross-attention model **423** of the image restoration model. F′″.sub.p of Equation 10 may indicate feature information outputted from the multi-head cross-attention model **423**.

[00009][Equation10]$F_p''' = \text{LN}(\text{softmax}(\frac{Q_v K_p''^T}{\sqrt{d}})V_p'')$

[0079] A query for performing multi-head cross-attention of Equation 10 may correspond to the feature information custom-character.sup.c×hw of the shallow CNN **421**. d of Equation 10 may indicate a dimension of a key vector. Q.sub.v of Equation 10, which is a projection (e.g., a projection based on an fc layer) of F.sub.v of Equation 5, may indicate a query vector. K″.sub.p and V″.sub.p, which are projections (e.g., the projection based on the fc layer) of F.sub.p″ of Equation 9, may indicate a key vector and a value vector, respectively. LN of Equation 10 may indicate a linear computation. Q.sub.vK″.sub.p.sup.T computation of Equation 10 may indicate an attention score of self-attention. T computation of Equation 10 may indicate a matrix transpose computation.

[0080] A key and a value for performing the multi-head cross-attention of Equation 10 may have a size of custom-character.sup.l×c (e.g., K″.sub.p∈custom-character.sup.l×c). It may indicate feature dimension (the number) of the shallow CNN **421** of custom-character.sup.l×c. Q.sub.v.Math.K″.sub.p.sup.T of Equation 10 may be=custom-character.sup.hw×l, and Q.sub.v.Math.K″.sub.p.sup.T.Math.V″.sub.p of Equation 10 may be custom-character.sup.hw×c. Referring to Equation 10, feature information F‴.sub.p obtained using a softmax computation and a layer normalization (LN) computation may be obtained from the multi-head cross-attention model **423**.

[0081] With respect to the feature information F‴.sub.p obtained from the multi-head cross-attention model **423**, the electronic device may perform calculations indicated by a chain connection of a merge model **424**, a first layer normalization model **425**, a feedforward model **426**, and a second layer normalization model **427**. Referring to FIG. **4**, a residual connection for an element-wise sum may be formed between the first layer normalization model **425** and the second layer normalization model **427**. The residual connection may be formed between the first layer normalization model **425** and the second layer normalization model **427** independently of the feed forward model **426**.

[0082] Referring to FIG. **4**, with respect to information obtained from the second layer normalization model **427**, the electronic device may repeatedly perform calculations based on a BiLSTM model **430** N times (e.g., 5 times). A combination of a first convolution model **428**, a second convolution model **429**, and the BiLSTM model **430** connected to the second layer normalization model **427** may be referred to as a decoder **470**. In an embodiment, feature information F obtained from the second layer normalization model **427** and to be inputted to the decoder **470** may be indicated as Equation 11.

[00010][Equation11]$F = LN(F_p''' \text{ .Math. } W_f + F_p''')$

[0083] Wf of Equation 11, which is an fc layer (or weights of the fc layer), may indicate a layer defined for a projection computation and a computation of the layer. The decoder **470** may have a structure (sequential-recurrent block, SRB) in which calculations indicated by the BiLSTM model **430** are repeatedly performed N times. The electronic device **101** may increase a resolution and/or a size of an image (e.g., an image indicated by the feature information F of Equation 11) outputted by the decoder **470** by using a pixel shuffle model **431**. For example, an output image **403** outputted from the pixel shuffle model **431** of the image restoration model may be determined based on Equation 12.

[00011][Equation12]RestoredImage = PixelSuffle(SRB(F_v, F))

[0084] When training the image restoration model having the structure of FIG. **4** (e.g., a second step of a training process), a loss function to be used for training the image restoration model may indicate a difference between a truth image corresponding to the input image **402** and the output image **403**. For example, a L1 distance (e.g., Manhattan distance and/or rectangular street grid) between the truth image and the output image **403** may be determined as the loss function. An embodiment is not limited thereto, and a L2 distance (or mean squared loss), a structural similarity index (SSIM), a triplex SSIM (TSSIM), and a Kullback-Leibler (KL) Divergence loss function for knowledge distillation may be used. For example, a loss function custom-character.sub.s based on the L2 distance may be defined as in Equation 13.

[00012][Equation13]$\mathcal{L}_s = \text{ .Math. } I_{SR} - I_{HR} \text{ .Math. }_2$

[0085] I.sub.SR of Equation 13 may indicate the output image **403**, and I.sub.HR may indicate a truth image. For training the image restoration model based on structural information of text, a loss function based on the TSSIM may be used, for example, such as a loss function custom-character.sub.tssim of Equation 14.

[00013][Equation14]$\mathcal{L}_{tssim} = 1 - \text{TSSIMsuchthatTSSIM} = \frac{(\ _x\ _y + \ _y\ _z + \ _x\ _z + C_1)(\ _{xy} + \ _{yz} + \ _{xz} + C_2)}{(\ _x^2 + \ _y^2 + \ _z^2 + C_1)(\ _x^2 + \ _x^2 + \ _x^2 + C_2)}$

[0086] In Equation 14, x may correspond to the degraded output image **403**, y may correspond to the output image **403**, and z may correspond to a truth image. Each of μ and σ of Equation 14 is a mean and standard deviation of corresponding images (e.g., x, y, and z). C of Equation 14 may be an epsilon value (e.g., a preset number set to prevent a zero division error).

[0087] According to an embodiment, the electronic device may perform training on the image restoration model by using the pre-trained sub-model **220**. The image restoration model may include the TPS **420** and the shallow CNN **421**, and may include an encoder for extracting feature information from the input image **402**. The image restoration model may include a fusion layer (e.g., the multi-head cross-attention model **423**) to combine implicit information of an intermediate layer prior to an output layer of the sub-model **220** which receives the input image **402** and the feature information. The image restoration model may include a decoder (e.g., the combination of the first convolution model **428**, the second convolution model **429**, and the BiLSTM model **430**), that is connected to the fusion layer, to generate the output image **403** having a second resolution greater than a first resolution of the input image **402**. The trained image restoration model may be provided as a portion of a software application (e.g., the image restoration program **125** of FIG. **1**) to restore an image.

[0088] Hereinafter, an exemplary structure of an image restoration model connected to the teacher model **220** of FIG. **2** and/or FIG. **3** will be described with reference to FIG. **5**.

[0089] FIG. **5** illustrates an exemplary block diagram of an image restoration model connected to a teacher model **210**. The electronic device **101** and/or the processor **110** of FIG. **1** may obtain, generate, and/or train the image restoration model described with reference to FIG. **5** by executing an image restoration program **125**.

[0090] As described above with reference to FIG. **3**, output data of a sub-model **220** may include a projection of output data of an RNN decoder (e.g., the RNN decoders **350** of FIG. **3**) as in Equation 2. An input of an NCAP projector **410** may include entire intermediate state vectors (e.g., hidden state vector) of the RNN decoder in each of the plurality of timings. In case that the sub-model **220** performs parallel decoding, the input of the NCAP projector **410** may include entire feature information obtained by the parallel decoding.

[0091] Output data of the teacher model **210** receiving an image **501** may be indicated as in Equation 15.

[00014][Equation15]$t_{\text{HR}} = \text{STR}_{\text{tea, dec}}(\text{STR}_{\text{tea, enc}}(x_{\text{LR}}))$ .Math. $W_{c,\text{HR}}$

[0092] The output data of the sub-model **220** may have a relationship of Equation **16**. t.sub.HR of Equation 15 may indicate an output of the teacher model **210** to which a high-resolution image is inputted. For example, t.sub.HR, which is information sequentially processed by an encoder and a decoder of an STR, may indicate information (e.g., probability distribution of text) projected by an fc layer. For example, W.sub.c of Equation 15 may indicate an fc layer, and x.sub.LR may indicate a low-resolution image.

[00015][Equation16]$t_{LR} \in \mathbb{R}^{1\times \text{ .Math. } A \text{ .Math.}} = \text{STR}_{\text{stu, dec}}(\text{STR}_{\text{stu, enc}}(x_{\text{LR}}))$ .Math. $W_{c,\text{LR}}$

[0093] t.sub.LR of Equation 16 may indicate an output of the sub-model **220** to which a low-resolution image is inputted. For example, Equation **16** may indicate output data of the sub-model **220**, which receives an image **502**.

[0094] Based on implicit information obtained from the sub-model **220**, the electronic device may obtain p.sub.NCAP of Equation 6 from the NCAP projector **410**.

[0095] When training the sub-model **220**, the electronic device may use a loss function custom-character .sub.distill of Equation 17 to reduce a domain gap (e.g., a domain difference between a high-resolution output image **503** and the low-resolution input image **502**) of prior knowledge of the sub-model **220**.

[00016][Equation17] $\mathcal{L}_{\text{distill}} = \text{ .Math. } t_{\text{HR}} - t_{\text{LR}} \text{ .Math. }_{1} + D_{\text{KL}}(t_{\text{LR}} \text{ .Math. } \text{ .Math. } t_{\text{HR}})$ (1)

$\mathcal{L}_{\text{distill}}(t_{\text{LR}}(\ ), t_{\text{HR}}(\ )) = \text{ .Math. }^{2} \text{ .Math.}_{i}\, t_{\text{HR},i}(\ ) \log\frac{t_{\text{HR},i}(\ )}{t_{\text{LR},i}(\ )} L_{\text{distill}}$ (2)

[0096] Equation (2) of Equation 17 may be a loss function based on a profile that is smooth (e.g., smoothness of kl divergence) based on temperature scaling, and Equation (1) of Equation 17 may be a loss function based on a profile that is sharp (e.g., smoothness of kl divergence) based on |t.sub.HR−t.sub.LR|.sub.1. The electronic device may determine the loss function custom-character.sub.distill using any one of the two equations of Equation 17. t.sub.HR of Equation 17 and t.sub.LR of Equation 17 may indicate prior knowledge obtained by inputting each of a high-resolution image and a low-resolution image to the STR including the sub-model **220**. t.sub.HR of Equation 17 may be generated from the frozen teacher model **210**. t.sub.LR of Equation 17 may be generated from the sub-model **220** of a trainable state. The loss function of Equation 17 may be determined by another method (e.g., L1 distance). Referring to Equation 17, truth data t.sub.HR(τ) of a soft label may be used to determine the loss function custom-character.sub.distill. Additionally, truth data t.sub.HR may be used to generate a smoother profile. β and τ of Equation 17 are parameters for controlling a soft label and smoothness, and may be set to, for example, 0.7 and 5, respectively.

[0097] A loss function custom-character.sub.str of the sub-model **220** may be determined by a loss function custom-character.sub.aux of Equation 18, which maximizes cross entropy loss (e.g., cross entropy of Equation 3) and a margin with a truth image, as in Equation 18. For example, in order to use a hard label for training, any one of the two equations of Equation 18 may be used.

[00017][Equation18] $\mathcal{L}_{\text{CE}}(t_{\text{LR}}, y_{\text{gt}}) = -\text{ .Math. } y_{\text{gt},i} \log t_{\text{LR},i}$ (1) $\mathcal{L}_{\text{str}} = \mathcal{L}_{\text{CE}}(p_{\text{pred}}, y_{\text{gt}}) + \mathcal{L}_{\text{aux}}$ (2)

[0098] When Equation (1) of Equation 18 is used, cross entropy loss custom-character.sub.CE used in scene text recognition may be used. When Equation (2) of Equation 18 is used, an additional loss function custom-character .sub.aux that maximizes a second margin may be used. y.sub.gt of Equation 18 may indicate a correct answer label (e.g., ground truth data) for an image received as an input. In an exemplary embodiment of FIG. **5**, y.sub.gt may be “recycled”.

[0099] The loss function custom-character.sub.aux of Equation 18 may correspond to custom-character.sub.aux of Equation 4. With respect to the sub-model **220** trained by a loss function custom-character.sub.str of Equation 18, the electronic device may further apply a loss function that reduces a difference between an attention score of a high-resolution image and a low-resolution image and/or a loss function that reduces a difference between probability distribution of the high-resolution image and probability distribution of the low-resolution image. For example, the electronic device may use a loss function to focus on an area associated with one or more characters within the input image **502**. For example, the electronic device may increase a weight for a string that may be confused by using weighted cross entropy (WCE), such as Equation 19.

[00018][Equation19]$\mathcal{L}_{txt}$ = .Math. .Math. A_{HR} - A_{SR} .Math. $_{1}$ + .Math. WCE(p_{pred}, y_{gt})

[0100] For example, in Equation 19, α may be set to a numerical value such as 10, and β may be set to a numerical value such as 0.0005. ‖A.sub.HR−A.sub.SR‖.sub.1 of Equation 19 may mean an L1 distance. Each of A.sub.HR and A.sub.SR of Equation 19 may indicate attention information (e.g., an attention map) of each of a high-resolution image and a low-resolution image. p.sub.pred of Equation 19 may indicate the output image **503**. A loss function custom-character.sub.txt of Equation 19 may be defined to reduce a difference between the attention information A.sub.SR for the low-resolution image and the attention information A.sub.HR for the high-resolution image.

[0101] In an embodiment, the electronic device may at least partially train the image restoration model using a combination of the loss functions exemplified above (e.g., joint learning). A combination custom-character .sub.total of loss functions may be set as in Equation 20. Using custom-character.sub.total of Equation 20, which is an example of the WCE, backpropagation of an entire model, starting from a pixel shuffle model **431**, may be performed. The backpropagation may be performed to reduce an error in the attention map and a logit. For example, the loss function custom-character.sub.txt of Equation 19 may be used to reduce an error between the attention map and text logit information obtained from an additional artificial intelligence model (e.g., text recognition network) to process the image **502**. By the backpropagation, the entire image restoration model may be trained to reduce a difference between the image **403** and the image **502**.

[00019][Equation20]$\mathcal{L}_{total}$ = $_{1}\mathcal{L}_{s}$ + $_{2}\mathcal{L}_{tssim}$ + .Math. $\mathcal{L}_{distill}$ + (1 -) .Math. $\mathcal{L}_{str}$ + $_{3}\mathcal{L}_{txt}$

[0102] Numerical values such as λ.sub.1=1, λ.sub.2=1, λ.sub.3=0.01 α=0.5 of Equation 20 may be set. α of Equation 20 may be a parameter for adjusting a training ratio between custom-character.sub.distill of Equation 17 and custom-character.sub.str of Equation 18. An embodiment is not limited thereto. custom-character.sub.s of Equation 20 may be determined as in Equation 13. custom-character.sub.tssim of Equation 20 may be defined as in Equation 10. custom-character.sub.distill of Equation 20 may be defined as in Equation 17. custom-character .sub.str of Equation 20 may be defined as in Equation 18. custom-character.sub.txt of Equation 20 may be defined as in Equation 19.

[0103] According to an embodiment, the electronic device may execute the image restoration model including the sub-model **220** and the projection model **230**, which may be executed at least temporarily simultaneously with models **510** for restoring an input image **502**. The models **510** may be combined with any sub-model **220** for recognizing a character that has been pre-trained. Using the sub-model **220**, the electronic device may effectively obtain prior knowledge (or prior information) to be used to restore or enhance the input image **502**.

[0104] Hereinafter, a performance of the image restoration model configured to obtain the output image **503** from the input image **502** will be described with reference to FIGS. **6** to **7**.

[0105] FIG. **6** illustrates graphs **611**, **612**, **621**, and **622** for describing a performance of an electronic device executing an image restoration model according to an embodiment. The graphs of FIG. **6** may be empirical graphs indicating the performance of the image restoration model executed by the electronic device and/or the image restoration program of FIGS. **1** to **5**.

[0106] Referring to FIG. **6**, the graph **611** indicating a mean of the number of Top 5 predictions of characters (e.g., numbers from 0 to 9 and alphabets from a to z) and the graph **612** indicating the number of Top 1 predictions are illustrated. Referring to FIG. **6**, the graph **621** indicating a standard deviation of the number of Top 5 predictions of characters and the graph **622** indicating a standard deviation of the number of Top 1 predictions are illustrated. Since an increase in a standard deviation means that there are fewer values around an mean value, the increase in the standard deviation may mean that accuracy of recognizing a character from an image is increased. For example, Table 1 may indicate a mean and a standard deviation of a result of predicting a character obtained by executing a sub-model (e.g., the sub-model **220** of FIG. **2**) with respect to all characters.

TABLE-US-00001 TABLE 1 Mean Std Top5 Top1 Total Top5 Baseline 448.583 408.944 67.048 159.964 Ours 484.028 435.056 71.161 169.166

[0107] “Ours” of Table 1 may indicate a mean and a standard deviation of a predicted result by executing the image restoration model according to an embodiment, and baseline of Table 1 may indicate a mean and a standard deviation of a predicted result by executing another model different from the image restoration model according to an embodiment.

[0108] FIG. **7** illustrates graphs **711**, **712**, **721**, **722**, **731**, **732**, **741**, and **742** for describing a performance of an electronic device executing an image restoration model according to an embodiment. Referring to FIG. **7**, areas **710**, **720**, **730**, and **740** corresponding to each of preset characters (e.g., 5, 9, u, and k) from a plurality of images are illustrated. For example, in the area **710**, the graph **712** indicating a mean of Top 5 predicting a preset character ‘5’ and the graph **711** indicating a baseline are illustrated. For example, in the area **720**, the graph **722** indicating a mean of Top 5 predicting a preset character ‘9’ and the graph **721** indicating a baseline are illustrated. For example, in the area **730**, the graph **732** indicating a mean of Top 5 predicting a preset character ‘u’ and the graph **731** indicating a baseline are illustrated. For example, in the area **740**, the graph **741** indicating a mean of Top 5 predicting a preset character ‘k’ and the graph **742** indicating a baseline are illustrated. Referring to the areas **710**, **720**, **730**, and **740**, in a result of predicting each of characters, a predicted value of a maximum predicted character may have a high value, and a frequency of predicting (e.g., confusing) a character less than or equal to Top 2 may be reduced.

[0109] For example, Table 2 may include a mean and a standard deviation of a result of predicting the preset characters using the sub-model.

TABLE-US-00002 TABLE 2 Top5 Mean Top1 Mean 5 9 k u 5 9 k u Baseline 222 87 166 424 133 63 137 382 Ours 229 92 183 482 206 80 156 429

[0110] For example, Table 3 may include the standard deviation of the result of predicting the preset characters using the sub-model.

TABLE-US-00003 TABLE 3 All Std Top5 Std 5 9 k u 5 9 k u Baseline 24.682 10.426 22.389 62.459 52.278 23.174 51.963 148.607 Ours 33.690 13.012 25.523 70.085 80.111 30.800 59.741 166.338

[0111] In order to check whether prior knowledge generated by the sub-model is biased, the electronic device may calculate a relationship between prior knowledge accuracy and STISR accuracy. The relationship may use Pearson Correlation Coefficient of Equation 21.

[00020][Equation21] $$_{XY} = \frac{.\text{Math.}_i^n\ (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{.\text{Math.}_i^n\ (X_i - \bar{X})^2}\sqrt{.\text{Math.}_i^n\ (Y_i - \bar{Y})^2}}$$

[0112] X of Equation 21 may indicate an output of the sub-model and/or a word error rate (WER) of a logit. X of Equation 21 may be defined as the WER and a CER of text logits of a student recognizer. Y of Equation 21 may be defined as Y=STISR WER and CER. The CER may be an error rate of a character (e.g., a character error rate). n of Equation 21 may indicate the number of total data, and i may indicate an index defined to perform a sum computation.

[0113] In an embodiment, Table 4 may indicate a Pearson relationship between prior knowledge and STISR accuracy.

TABLE-US-00004 TABLE 4 Prior SR Pearson Error Rate Error Rate Correlation Method WER CER WER CER WER CER TATT 52.3% 32.2% 47.2% 30.7% 0.7146 0.8026 TATT 37.4% 21.3% 43.3% 27.1% 0.6626 0.7359 w/Ours Δ −14.9% −11.0% −3.9% −3.6% −7.3% −8.3% LEMMA 76.1% 58.3% 44.0% 28.3% 0.3465 0.4580 LEMMA 77.6% 60.5% 42.1% 26.9% 0.3279 0.3052 w/Ours Δ +1.5% +2.2% −1.95% −1.34% −5.4% −33.4%

[0114] Referring to Table 4, compared to a conventional method (baseline), the Pearson Correlation Coefficient of the image restoration model (e.g., Ours) according to an embodiment may be relatively reduced in both the WER and the CER. The Pearson Correlation Coefficient of the image restoration model being reduced may mean that the image restoration model is not dependent on incomplete information (e.g., prior knowledge).

[0115] In an embodiment, Table 5 may indicate a relationship between performance improvement of the electronic device and a parameter increase amount.

Table 5

[0116] Referring to Table 5, when a parameter of the image restoration model is increased by approximately 0.3%, performance improvement may be expected. According to an embodiment, the electronic device may use a commonly used adapter (e.g., multi-layer perceptron (MLP)) and/or a convolution type adapter to execute the image restoration model. In an embodiment, when a 1×1 convolution type adapter is used, the performance may be relatively further improved.

[0117] As described above, the electronic device according to an embodiment may execute the image restoration model configured to generate text-related information (e.g., a text probability map) from an image. The image restoration model may include the sub-model that is pre-trained to generate the information from the image. The image restoration model may restore or enhance the image using implicit information used to generate explicit information (e.g., one or more characters associated with an image, and a relative position of the one or more characters) outputted from the sub-model. Since an image is restored using information associated with text, the electronic device may be trained to interpret a license plate and/or a sign plate.

[0118] Hereinafter, license plates restored by the image restoration model are exemplarily illustrated with reference to FIGS. **8**A and/or **8**B.

TABLE-US-00005 Method NCAP Adapters MACs #Params TATT 4.60 G 31.44 M TATT w/Ours ✓ 4.64 G 31.52 M ✓ ✓ 4.43 G 31.52 M Δ −3.7% +0.3% LEMMA 6.69 G 39.75 M LEMMA w/Ours ✓ 6.69 G 39.90 M ✓ ✓ 6.71 G 39.90 M Δ +0.3% +0.4%

[0119] FIGS. **8**A and **8**B illustrate at least one license plate (or number plate), which is a subject included in an image restored by an image restoration model according to an embodiment.

[0120] Referring to FIG. **8**A, images **810** including at least one license plate obtained from the image restoration model are illustrated. The images **810** may be outputted from, or provided by, an electronic device that executes the image restoration model as a result of restoring or enhancing a low-resolution input image (e.g., the input image **202** of FIG. **2**).

[0121] For example, the electronic device may generate an image **820** including a license plate based on the law of the Republic of Korea. The image **820** may include numbers (e.g., 12) indicating a type of a vehicle, an alphabet (e.g., "custom-character") indicating a purpose of the vehicle, and numbers (e.g., 1234) indicating a serial number uniquely assigned to the vehicle. For example, the electronic device may obtain an image **830** including a license plate based on the law of the Republic of Korea. The image **830** may further include, with respect to the image **820**, characters (e.g., a place name such as "Seoul") indicating an area associated with the license plate. A background color of the license plate represented through the images **820** and **830** may indicate a category (e.g., a private vehicle) of the vehicle defined by the law of the Republic of Korea.

[0122] For example, the electronic device may generate an image **840** including a license plate based on the law of China. In the image **840**, a character (e.g., custom-character) indicating an area associated with the license plate and a character (e.g., N) indicating a city (e.g., a sub-area of the area) associated with the license plate may include information on the area or purpose. The image **840** may include serial numbers (e.g., 888R8) uniquely assigned to a vehicle. A color of the license plate represented through the image **840** may indicate a category (e.g., a passenger car, a large vehicle, a bus, a truck, and/or a motorcycle) of the vehicle.

[0123] For example, the electronic device may generate an image **850** including a license plate based on the law of the European Union. The image **850** may include a symbol indicating the European Union, characters (e.g., EST) indicating an area associated with the license plate, and serial numbers (e.g., "307 RTB") uniquely assigned to a vehicle on which the license plate is mounted. An embodiment is not limited thereto, and the image **850** may further include a flag of a country in which the vehicle on which the license plate is registered as a country affiliated with the European Union.

[0124] For example, the electronic device may generate an image **860** including a license plate based on the law of Japan. The image **860** may include characters (e.g., custom-charactercustom-character) indicating an area, numbers (e.g., 500) indicating a category of a vehicle, a character indicating a purpose of a business associated with the vehicle, and serial numbers (e.g., 46-49) uniquely assigned to the vehicle on which the license plate is mounted.

[0125] Referring to FIG. **8**B, images **870** including a license plate based on the law of the United States generated by the electronic device according to an embodiment are illustrated. Referring to the images **870**, based on the law of the United States, the license plate including an image and/or a figure defined by a state government of the United States may be generated. The license plate may include text (e.g., "TEXAS", "ALABAMA", "KENTUCKY", and the like) indicating a state government together with an image and/or a figure indicating the state government in which a vehicle is registered. Together with the text, the image representing the license plate may include a serial number (e.g., a combination of alphabets and/or numbers such as "GV71P") uniquely assigned to the vehicle.

[0126] FIG. **9** is a diagram for describing an overconfidence phenomenon. When training of the super-resolution network **241**, **242**, **243**, and **244** of FIG. **2** and training of a character recognition network (e.g., the sub-model **220** of FIG. **2**) are performed simultaneously, an overconfidence phenomenon may occur due to a difference in training speed between the super-resolution network and the character recognition network. The overconfidence phenomenon may include a phenomenon of predicting an incorrect character with a high probability value from an image including a character that is difficult to infer. The overconfidence phenomenon may negatively affect a result (e.g., character probability distribution) of the character recognition network (e.g., the sub-model **220** of FIG. **2**). According to an embodiment, an electronic device may reduce the overconfidence phenomenon by using a loss function that combines hard-level truth data and a soft label (e.g., output of a teacher model), as described above with reference to Equation 17 and Equation 18.

[0127] Referring to FIG. **9**, graphs **910** and **920** indicating reliability at each of a word level and a character level are illustrated. An x-axis of the graphs **910** and **920** may indicate a probability value corresponding to a character recognized from a neural network. A y-axis of the graphs **910** and **920** may indicate a probability (e.g., accuracy) that a character recognized by the neural network is a correct answer. As the probability value and the accuracy are more proportional, it may be explained that the overconfidence phenomenon is reduced, as in baselines **919** and **929** of the graphs **910** and **920**.

[0128] A line **911** of the graph **910** may indicate an ideal relationship between accuracy and reliability of an image restoration model of the electronic device. A line **912** of the graph **910** may indicate accuracy of the image restoration model trained based on a soft label. Lines **913** of the graph **910** may indicate accuracy of the image restoration model trained based on a hard label. A line **921** of the graph **920** may indicate an ideal relationship between accuracy and reliability of the image restoration model of the electronic device. A line **922** of the graph **920** may indicate the accuracy of the image restoration model trained based on a soft label. Lines **923** of the graph **920** may indicate accuracy of the image restoration model trained based on a hard label. When trained with only the hard label, the accuracy may be reduced compared to a probability value. When trained with only the soft label, the overconfidence phenomenon is reduced, but a performance may be degraded. Referring to the graphs **910** and **920** of FIG. **9**, the lines **911** and **921** indicating the accuracy of the image restoration model of the electronic device according to an embodiment may be positioned closer to the baselines **919** and **929** (e.g., a baseline indicating accuracy of the image restoration model with minimized overconfidence phenomenon) indicating ideal accuracy than the other lines **912**, **913**, **922**, and **923**.

[0129] In an embodiment, a method of increasing or enhancing a resolution of an image in which one or more characters are captured using a model trained to output explicit information such as a text probability map may be required. In an embodiment, a method of increasing or enhancing the resolution of the image in which one or more characters are captured using implicit information of an intermediate layer in the model trained to output the explicit information may be required. As described above, according to an embodiment, a non-transitory computer readable storage medium storing instructions may be provided. The instructions, when executed by at least one processor of an electronic device individually or collectively, may cause the electronic device to receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on the received request, execute an image restoration model including, an encoder to extract feature information from the first image, a sub-model to determine a text-probability map with respect to the first image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information, and a decoder to generate an image of the second resolution, which is connected to the fusion layer. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model. According to an embodiment, the electronic device may increase or enhance the resolution of the image in which one or more characters are captured using a model trained to output explicit information such as the text probability map. According to an embodiment, the electronic device may increase or enhance the resolution of the image in which one or more characters are captured by using the implicit information of the intermediate layer in the model trained to output the explicit information.

[0130] For example, the instructions, when executed by the at least one processor of the electronic device individually or collectively, may cause the electronic device to execute the image restoration model including the fusion layer, which is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.

[0131] For example, the sub-model may be trained to output the text probability map indicating one or more characters indicated as being captured by the first image, and positions of the one or more characters.

[0132] For example, the sub-model may be pre-trained by a teacher model, the teacher model is executed using parameters more than parameters for the sub-model.

[0133] For example, the instructions, when executed by the at least one processor of the electronic device individually or collectively, may cause the electronic device to receive, from an external electronic device through communication circuitry of the electronic device, a first signal including the request and a third image. The instructions, when executed by the at least one processor of the electronic device individually or collectively, may cause the electronic device to, based on receiving the first signal, segment, within the third image, a portion associated with a license plate as the first image. The instructions, when executed by the at least one processor of the electronic device individually or collectively, may cause the electronic device to, based on obtaining the second image from the image restoration model executed using the segmented first image, transmit a second signal including the second image to the external electronic device.

[0134] As described above, according to an embodiment, an electronic device may comprise memory storing instructions, and at least one processor configured to execute the instructions. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on the received request, execute an image restoration model including an encoder to extract feature information from the first image, a sub-model to determine a text-probability map with respect to the first image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information, and a decoder to generate an image of the second resolution, which is connected to the fusion layer. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.

[0135] For example, the instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to execute the image restoration model including the fusion layer, which is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit

information.
[0136] For example, the sub-model may be trained to output the text probability map indicating one or more characters indicated as being captured by the first image, and positions of the one or more characters.
[0137] For example, the sub-model may be pre-trained by a teacher model, the teacher model is executed using parameters more than parameters for the sub-model.
[0138] For example, the instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to receive, from an external electronic device through communication circuitry of the electronic device, a first signal including the request and a third image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on receiving the first signal, segment, within the third image, a portion associated with a license plate as the first image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on obtaining the second image from the image restoration model executed using the segmented first image, transmit a second signal including the second image to the external electronic device.
[0139] As described above, according to an embodiment, a method of an electronic device may be provided. The method may comprise, based on receiving an image, obtaining a sub-model trained to output a text-probability map indicating one or more characters associated with the image. The method may comprise performing, using the sub-model, training of an image restoration model including an encoder to extract feature information from an input image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, prior to an output layer of the sub-model which receives the input image, and the feature information, and a decoder, that is connected to the fusion layer, to generate an output image having a second resolution greater than a first resolution of the input image. The method may comprise providing the image restoration model as a portion of a software application to restore the image.
[0140] For example, the image restoration model may include the fusion layer that is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.
[0141] For example, the sub-model may be trained to output the text probability map indicating one or more characters indicated as being captured by the input image, and positions of the one or more characters.
[0142] For example, the obtaining may comprise obtaining the sub-model using a teacher model that is executed using parameters more than parameters for the sub-model.
[0143] For example, the providing may comprise, in response to a request to restore a portion associated with a license plate segmented from a source image, executing the image restoration model.
[0144] As described above, according to an embodiment, an electronic device may comprise memory storing instructions, and at least one processor configured to execute the instructions. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, based on receiving an image, obtain a sub-model trained to output a text-probability map indicating one or more characters associated with the image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to perform, using the sub-model, training of an image restoration model including an encoder to extract feature information from an input image, a fusion layer to combine implicit information of an intermediate layer of the sub-model, prior to an output layer of the sub-model which receives the input image, and the feature information, and a decoder, that is connected to the fusion layer, to generate an output image having a second resolution greater than a first resolution of the input image. The instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to provide the image restoration model as a portion of a software application to restore the image.
[0145] For example, the image restoration model may include the fusion layer that is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.
[0146] For example, the sub-model may be trained to output the text probability map indicating one or more characters indicated as being captured by the input image, and positions of the one or more characters.
[0147] For example, the instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to obtain the sub-model using a teacher model that is executed using parameters more than parameters for the sub-model.
[0148] For example, the instructions, when executed by the at least one processor individually or collectively, may cause the electronic device to, in response to a request to restore a portion associated with a license plate segmented from a source image, execute the image restoration model.
[0149] The above-described device may be implemented as hardware components, software components, and/or a combination of hardware components and software components. For example, the devices and components described in the embodiments may be implemented using one or more general-purpose computers or special-purpose computers, such as e.g., a processor, a controller, an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a programmable logic unit (PLU), a microprocessor, or any other device capable of executing and responding to instructions. The processing device may perform an

operating system (OS) and one or more software applications performed on the operating system. Further, the processing device may access, store, manipulate, process, and generate data in response to the execution of the software. For convenience of understanding, it may be described that one processing device is used. However, those skilled in the art may understand that the processing device may include a plurality of processing elements and/or a plurality of types of processing elements. For example, the processing device may include a plurality of processors or one processor and one controller. In addition, other processing configurations such as parallel processors are also possible.

[0150] The software may include a computer program, a code, an instruction, or one or more combinations thereof, and may configure a processing device to operate as desired or may independently or collectively instruct the processing device. Software and/or data may be interpreted by a processing device or may be embodied in any type of machine, component, physical device, computer storage medium, or device to provide a command or data to the processing device. Software may be distributed on a networked computer system and stored or executed in a distributed manner. Software and data may be stored in one or more computer-readable recording media.

[0151] The method according to an embodiment of the disclosure may be implemented in the form of program commands executable by various computer means and recorded on a computer-readable medium. In this case, the medium may be a persistent storage of a computer-executable program, or it may be a temporary storage for execution or download. Further, the medium may be various recording means or storage means in which a single piece of hardware or a plurality of pieces of hardware are combined, and the medium is not limited to a medium directly connected to a computer system, and may be distributed on a network. Examples of the medium may include a magnetic medium such as a hard disk, a floppy disk, and a magnetic tape, an optical recording medium such as a compact disc read only memory (CD-ROM) and a digital versatile disc (DVD), a magneto-optical medium such as a floptical disk, and a read only memory (ROM), a random access memory (RAM), a flash memory, etc. configured to store program instructions. In addition, examples of other media include recording media or storage media managed by an application store that distributes applications, a site that supplies or distributes various other software, a server, and the like.

[0152] As described above, although the embodiments have been described with reference to limited embodiments and drawings, various modifications and modifications may be made from the above description by those skilled in the art. For example, even if the described techniques are performed in a different order from the described method, and/or components such as the described system, structure, device, circuit, etc. are combined or combined in a different form from the described method, or are replaced or substituted by other components or equivalents, appropriate results may be achieved.

[0153] Therefore, other implementations, other embodiments, and those equivalent to the scope of the patent claim also fall within the scope of the patent claims to be described later.

Claims

1. A non-transitory computer readable storage medium storing instructions, wherein the instructions, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to: receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution; based on the received request, execute an image restoration model including: an encoder to extract feature information from the first image; a sub-model to determine a text probability map with respect to the first image; a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information; and a decoder to generate an image of the second resolution, which is connected to the fusion layer, provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.

2. The non-transitory computer readable storage medium of claim 1, wherein the instructions, when executed by the at least one processor of the electronic device individually or collectively, cause the electronic device to: execute the image restoration model including the fusion layer, which is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.

3. The non-transitory computer readable storage medium of claim 1, wherein the sub-model is trained to output the text probability map indicating one or more characters indicated as being captured by the first image, and positions of the one or more characters.

4. The non-transitory computer readable storage medium of claim 3, wherein the sub-model is pre-trained by a teacher model, the teacher model is executed using parameters more than parameters for the sub-model.

5. The non-transitory computer readable storage medium of claim 1, wherein the instructions, when executed by the at least one processor of the electronic device individually or collectively, cause the electronic device to: receive, from an external electronic device through communication circuitry of the electronic device, a first signal including the request and a third image; and based on receiving the first signal, segment, within the third image, a portion

associated with a license plate as the first image.
6. The non-transitory computer readable storage medium of claim 5, wherein the instructions, when executed by the at least one processor of the electronic device individually or collectively, cause the electronic device to: based on obtaining the second image from the image restoration model executed using the segmented first image, transmit a second signal including the second image to the external electronic device.
7. The non-transitory computer readable storage medium of claim 1, wherein the sub-model is further trained to generate implicit information to be used at the image restoration model including the sub-model, after being trained to output the text probability map.
8. An electronic device comprising: memory storing instructions; and at least one processor configured to execute the instructions, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: receive a request to restore a first image of a first resolution, to an image of a second resolution larger than the first resolution; based on the received request, execute an image restoration model including: an encoder to extract feature information from the first image; a sub-model to determine a text probability map with respect to the first image; a fusion layer to combine implicit information of an intermediate layer of the sub-model, which is positioned prior to an output layer trained to output the text probability map, and the feature information; and a decoder to generate an image of the second resolution, which is connected to the fusion layer, provide, as a response to the request, a second image of the second resolution that is obtained based on execution of the image restoration model.
9. The electronic device of claim 8, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: execute the image restoration model including the fusion layer, which is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.
10. The electronic device of claim 8, wherein the sub-model is trained to output the text probability map indicating one or more characters indicated as being captured by the first image, and positions of the one or more characters.
11. The electronic device of claim 10, wherein the sub-model is pre-trained by a teacher model, the teacher model is executed using parameters more than parameters for the sub-model.
12. The electronic device of claim 8, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: receive, from an external electronic device through communication circuitry of the electronic device, a first signal including the request and a third image; based on receiving the first signal, segment, within the third image, a portion associated with a license plate as the first image.
13. The electronic device of claim 12, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on obtaining the second image from the image restoration model executed using the segmented first image, transmit a second signal including the second image to the external electronic device.
14. The electronic device of claim 8, wherein the sub-model is further trained to generate implicit information to be used at the image restoration model including the sub-model, after being trained to output the text probability map.
15. A method of an electronic device, comprising: based on receiving an image, obtaining a sub-model trained to output a text probability map indicating one or more characters associated with the image; performing, using the sub-model, training of an image restoration model including: an encoder to extract feature information from an input image; a fusion layer to combine implicit information of an intermediate layer of the sub-model prior to an output layer of the sub-model which receives the input image, and the feature information; and a decoder, that is connected to the fusion layer, to generate an output image having a second resolution greater than a first resolution of the input image, and providing the image restoration model as a portion of a software application to restore the image.
16. The method of claim 15, wherein the image restoration model includes the fusion layer that is connected to the intermediate layer to extract the implicit information used to determine the text probability map which is explicit information.
17. The method of claim 15, wherein the sub-model is trained to output the text probability map indicating one or more characters indicated as being captured by the input image, and positions of the one or more characters.
18. The method of claim 15, wherein the obtaining comprises: obtaining the sub-model using a teacher model that is executed using parameters more than parameters for the sub-model.
19. The method of claim 15, wherein the providing comprises: in response to a request to restore a portion associated with a license plate segmented from a source image, executing the image restoration model.
20. The method of claim 15, wherein the performing the training comprises: further training the sub-model trained to output the text probability map using a loss function based on implicit information that is used by the image restoration model.